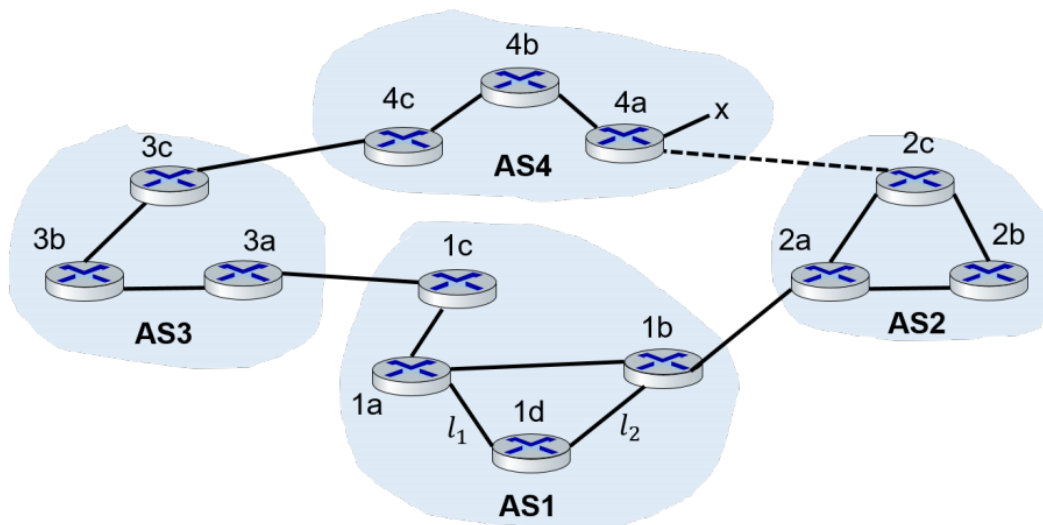


Problem Set 7

Issued: Mar. 30 Due: Jun. 5

Readings: Peterson - Computer Networks: A Systems Approach, (5th edition) Chapter 5

1. Consider the following network. Assume that AS1, AS2, AS3, and AS4 run intra-AS routing protocols OSPF. We also assume that this network adopts iBGP and eBGP as the inter-AS routing protocols.
- a) Which protocol is used by router 3c to get the prefix x? (OSPF, iBGP, or eBGP)
- b) Which protocol is used by router 3a to get the prefix x? (OSPF, iBGP, or eBGP)
- c) Which protocol is used by router 1c to get the prefix x? (OSPF, iBGP, or eBGP)
- d) Which protocol is used by router 1d to determine the path to 1c? (OSPF, iBGP, or eBGP)

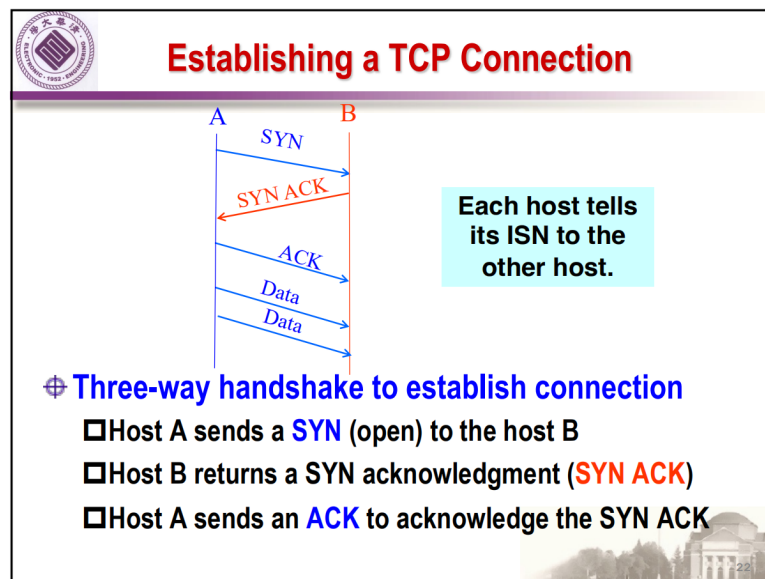


Solution: Through this problem, we want to make sure that you understand the difference between the intra-AS routing protocol and iBGP.

- a) eBGP
- b) iBGP
- c) eBGP
- d) OSPF

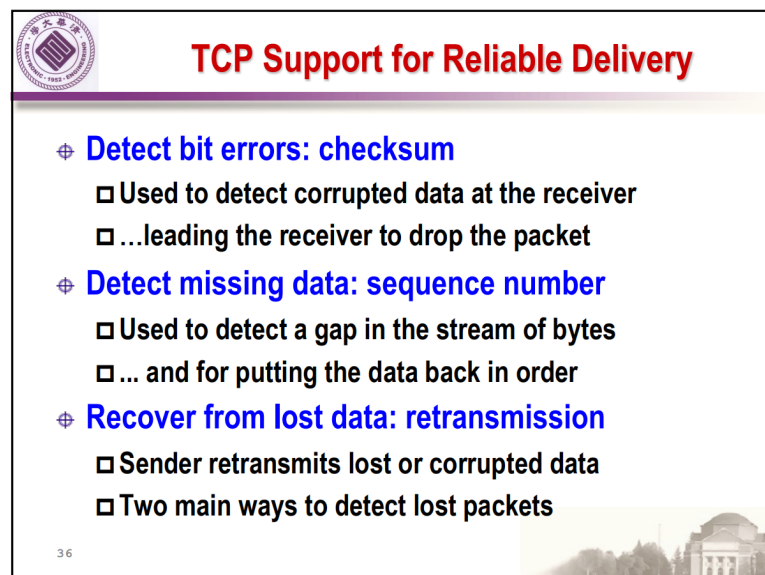
2. How to establish a TCP connection between host A and host B?

Solution:



3. List 3 mechanisms in TCP that are designed to support for reliable delivery. What are they used for?

Solution:



4. By adopting TCP, the cwnd of a host changes with round n as follow

a) Determine the periods during which TCP works in the slow start state and congestion avoidance state, respectively.

wnd	1	2	4	8	16	32	33	34	35	36	37	38	39
n	1	2	3	4	5	6	7	8	9	10	11	12	13
wnd	40	41	42	43	44	45	46	47	48	1	2	4	8
n	14	15	16	17	18	19	20	21	22	23	24	25	26

b) Determine when the 70th segment is sent.

Solution:

a) Slow start: 1~6, 23~26

Congestion avoidance: 7~22

b) Round 7